Planck-Boundary Geometry of Compact S^3 Cosmology

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We analyze a compact S^3 cosmological framework in the limiting regime $a \to \ell_P$. The Planck epoch is modelled as a regular self-adjoint boundary of minisuperspace, not as a classical curvature singularity: the effective domain is $a \in [\ell_P, \infty)$, the wave function may have a finite non-zero value at $a = \ell_P$, and the sub-Planck interval is not assigned a probability measure in the effective theory.

The model is restricted to the smooth closed connected orientable simply-connected spatial sector. In that sector, the Poincaré–Perelman theorem, together with uniqueness of smooth structures on three-manifolds, selects $\Sigma \simeq S^3$. The scalar spectrum of the round three-sphere is then organized by a Fibonacci coarse-graining. The golden ratio $\varphi = (1+\sqrt{5})/2$ enters only as the asymptotic ratio of successive Fibonacci scales; it is not an additional dynamical constant.

Within the stated assumptions we prove four conditional results. First, for a closed FLRW metric with $k = +1$, the Kretschmann scalar satisfies $K(t_P) \le C/\ell_P^4 < \infty$ whenever the regularised scale factor is bounded below and $H, \dot{H}$ are finite at the Planck boundary. This is a kinematic semiclassical regularity theorem, not a microscopic quantum-gravity derivation. Second, for any scale-covariant compact S^3 spectrum whose spectral zeta function is finite at the Casimir point,

$$\rho_{\rm vac}^{\rm reg}(a) = \frac{C_\zeta}{2\pi^2 a^4}.$$

For the conformally shifted scalar reference spectrum, $C_\zeta = 1/240$, so $\rho_{\rm vac}^{\rm reg} = 1/(480\pi^2 a^4)$. This is a regularised compact-spectrum contribution, not a prediction of the observed late-time dark energy density. Third, a self-adjoint Wheeler–DeWitt operator on $L^2([\ell_P, \infty), a\,da)$ gives zero probability flux through the Planck boundary. Fourth, diffeomorphism invariance gives Noether-II differential identities; in covariant phase space the full closed Cauchy slice has no Iyer–Wald boundary Hamiltonian variation because $\partial\Sigma = \varnothing$, although quasi-local charges on internal subdomains remain possible.

I. INTRODUCTION AND SCOPE

Classical FLRW cosmology, extrapolated without modification to $a = 0$, produces divergent curvature invariants. Perturbative quantum field theory, extrapolated without regularisation to arbitrarily short wavelengths, produces ultraviolet divergences. The Planck length

$$\ell_P = \sqrt{\hbar G/c^3}$$

marks the scale at which neither extrapolation should be taken as literal classical physics.

This paper isolates a conservative Planck-boundary structure. It does not claim to construct a full microscopic theory of quantum gravity. The framework has five ingredients:

1. a spatial sector assumption: smooth, closed, connected, orientable, and simply connected three-manifolds;
2. a closed FLRW homogeneous geometry with $k = +1$;
3. a lower effective boundary $a = \ell_P$ for minisuperspace;
4. a compact S^3 spectral zeta regularisation of vacuum sums;
5. a self-adjoint Wheeler–DeWitt boundary condition at $a = \ell_P$.

The mathematical question is then precise: what follows from these assumptions, and what does not?

The answer is deliberately limited. The model gives a conditional curvature bound, a finite regularised compact-spectrum scale, norm preservation in the minisuperspace sector, and the expected Noether-II identities of generally covariant dynamics. It does not derive finite $H(t_P)$ or $\dot{H}(t_P)$ from first principles, does not predict the observed cosmological constant, and does not turn the Fibonacci hierarchy into an exact spacetime symmetry.

Conventions. We use natural units $c = \hbar = k_B = 1$ except when defining ℓ_P. The reduced Planck mass is $m_P = (8\pi G)^{-1/2}$, the metric signature is $(-+++)$, and S^3 denotes the unit three-sphere. The physical radius is $a(t)$.

II. TOPOLOGICAL AND SPECTRAL FRAMEWORK

A. Selection of the S^3 sector

The topology is not derived from normalisability of the wave function. It is a sector assumption.

Assumption II.1 (Closed simply-connected spatial sector). The admissible spatial slice is restricted to the smooth, closed, connected, orientable, simply-connected three-manifold sector:

$$\Sigma \text{ smooth, closed, connected, orientable,} \qquad \pi_1(\Sigma) = 0.$$

Theorem II.2 (Topological uniqueness in the assumed sector)**.** *Under Assumption II.1, the spatial slice is homeomorphic to S^3. In the smooth three-dimensional category used here, it is diffeomorphic to the standard S^3:*

$$\Sigma \simeq S^3.$$

Proof. The Poincaré–Perelman theorem states that every closed simply-connected topological three-manifold is homeomorphic to S^3 [1–3], building on Hamilton's Ricci-flow programme [4]. Moise's theorem gives the uniqueness, up to diffeomorphism, of smooth structures on topological three-manifolds [5]. Hence in the smooth category relevant for the metric formulation the assumed sector is the standard S^3 sector. □

Remark II.3. Compactness supports a global L^2 inner product and orientability supplies a global volume form. Simple connectivity excludes lens spaces and other closed topological sectors. These are assumptions of the present model; they are not deduced from the Wheeler–DeWitt equation alone.

B. Fibonacci coarse-graining of the S^3 spectrum

The scalar Laplacian on the unit S^3 has eigenvalues

$$-\Delta_{S^3} Y_{\ell mn} = \ell(\ell+2) Y_{\ell mn}, \qquad \ell = 0, 1, 2, \ldots,$$

with eigenspace dimension $(\ell+1)^2$.

Let

$$F_0 = 0, \qquad F_1 = 1, \qquad F_{n+1} = F_n + F_{n-1} \quad (n \geq 1)$$

be the Fibonacci sequence. To avoid the repeated cut point $F_1 = F_2 = 1$, define a strictly increasing cut sequence by

$$\mathcal{F}_0 = 0, \qquad \mathcal{F}_1 = 1, \qquad \mathcal{F}_n = F_{n+1} \quad (n \geq 2).$$

The non-empty Fibonacci spectral bands are

$$B_n = \{\ell \in \mathbb{N}_0 : \mathcal{F}_n \leq \ell < \mathcal{F}_{n+1}\}, \qquad n = 0, 1, 2, \ldots.$$

Thus $B_0 = \{0\}$, $B_1 = \{1\}$, $B_2 = \{2\}$, $B_3 = \{3, 4\}$, and so on.

For non-constant modes, a characteristic angular half-wavelength scale is

$$d_n(t) \sim \frac{\pi a(t)}{\mathcal{F}_n}, \qquad n \geq 1.$$

At the Planck boundary $a_{\text{eff}}(t_P) \simeq \ell_P$, this formula is a formal spectral scale. The effective theory does not claim operational access to spatial resolution below ℓ_P.

Proposition II.4 (Planck-boundary resolution of the stratification)**.** *Assume $a_{\text{eff}}(t_P) \geq \ell_P$ and assume that the effective theory assigns no operational support to sub-Planck spatial resolution. Then the Fibonacci bands remain a well-defined organization of the compact S^3 spectrum, but only bands associated with resolvable observables belong to the semiclassical description. Modes whose formal wavelength is below ℓ_P are part of the mathematical spectral tower, not independent resolved degrees of freedom of the effective Planck-boundary theory.*

Remark II.5. The golden ratio enters through

$$\lim_{n\to\infty} \frac{\mathcal{F}_{n+1}}{\mathcal{F}_n} = \varphi = \frac{1+\sqrt{5}}{2}.$$

This is a consequence of the Fibonacci recurrence. It is not a new local coupling, field, or physical constant.

C. Metric near the boundary

The closed FLRW metric is

$$ds^2 = -dt^2 + a(t)^2 \left[d\chi^2 + \sin^2\chi \, (d\vartheta^2 + \sin^2\vartheta \, d\varphi^2)\right].$$

In the homogeneous semiclassical regime, $a(t)$ is interpreted as an effective scale variable, for instance as an expectation value of a geometric operator in a suitable state. Quantum corrections are represented schematically by

$$g_{ij}(t) = g^{(0)}_{ij}(t) + \hbar \, \Delta g_{ij}(t),$$

where $g^{(0)}_{ij}$ is the round S^3 spatial metric with physical scale factor $a(t)$. This notation records a semiclassical expansion; it is not a microscopic quantisation prescription.

Assumption II.6 (Regular Planck-boundary semiclassical data). At the Planck boundary the effective scale factor and kinematic data satisfy

$$\begin{aligned} &a_{\text{eff}}(t_P) \geq a_{\min} \geq \ell_P, \\ &|H(t_P)| < \infty, \\ &|\dot{H}(t_P)| < \infty. \end{aligned}$$

Metric fluctuations are assumed bounded in the restricted sense that their contribution to a_{eff}, H, and $\dot{H}$ remains finite.

Remark II.7. The spatial metric does not become Euclidean. In the homogeneous $k = +1$ sector the background spatial geometry remains the round metric on S^3, regularised only by the non-zero effective radius.

III. KINEMATIC CURVATURE BOUND

For a closed FLRW metric with $k = +1$, the four-dimensional Kretschmann scalar is

$$\begin{aligned} K &= R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \\ &= 12\left[\left(\frac{\ddot{a}}{a}\right)^2 + \left(\frac{\dot{a}^2+1}{a^2}\right)^2\right] \\ &= 12\left[(\dot{H}+H^2)^2 + \left(H^2+\frac{1}{a^2}\right)^2\right]. \end{aligned}$$

Classically, $a \to 0$ together with uncontrolled H or $\dot{H}$ produces a curvature singularity. The present framework instead removes $a = 0$ from the effective configuration space.

Theorem III.1 (Kinematic curvature bound from regular Planck-boundary data)**.** *For a closed FLRW metric with $k = +1$, assume that at the Planck boundary*

$$a_{\rm eff}(t_P) = \alpha_P \ell_P, \qquad \alpha_P \geq 1,$$

and

$$|H(t_P)| < \infty, \qquad |\dot{H}(t_P)| < \infty.$$

Then

$$K(t_P) \leq \frac{C}{\ell_P^4} < \infty,$$

where

$$\begin{aligned} C &= 12\left[(q_P + h_P^2)^2 + \left(h_P^2 + \alpha_P^{-2}\right)^2\right], \\ h_P &= \ell_P H(t_P), \qquad q_P = \ell_P^2 \dot{H}(t_P). \end{aligned}$$

Proof. Substitution into the FLRW expression gives

$$K(t_P) = \frac{12}{\ell_P^4}\left[(q_P + h_P^2)^2 + \left(h_P^2 + \alpha_P^{-2}\right)^2\right].$$

The assumptions imply that α_P^{-1}, h_P, and q_P are finite. Therefore C is finite. □

Remark III.2. The theorem is not a dynamical singularity-resolution theorem. It states the exact kinematic consequence of regular boundary data. A complete quantum-gravity model would still have to explain why the finite values of $H(t_P)$ and $\dot{H}(t_P)$ are dynamically selected.

Corollary III.3. *Within any interval on which $a_{\rm eff} \geq \ell_P$ and $H, \dot{H}$ remain finite, the closed-FLRW curvature scalar K remains finite.*

Interpretation III.4 (No external Planck-boundary time). The Wheeler–DeWitt equation is a constraint equation, $\hat{H}\Psi = 0$, and does not contain an external time parameter. A semiclassical clock can emerge in a WKB/Born–Oppenheimer regime, but that approximation is not assumed at the Planck boundary. Thus statements about $\partial_t g_{ij}$ at $a \simeq \ell_P$ are clock- and gauge-dependent unless a specific clock reconstruction has been supplied.

IV. SPECTRAL ZETA REGULARISATION OF THE VACUUM SCALE

A. Scale-covariant compact spectra

For a free field on a compact spatial manifold the formal vacuum energy is

$$E_{\rm vac} = \frac{1}{2}\sum_k \omega_k.$$

On a round S^3 of physical radius a, a scale-covariant spectral family has

$$\omega_k(a) = \frac{\nu_k}{a},$$

where ν_k is the dimensionless spectrum on the unit sphere. Define

$$\zeta_\omega(s) = \sum_k \nu_k^{-s}, \qquad \zeta_\omega(s;a) = \sum_k \omega_k(a)^{-s} = a^s \zeta_\omega(s).$$

If the spectral zeta function admits a finite analytic continuation to $s = -1$, the zeta-regularised Casimir energy is

$$E_{\rm vac}^{\rm reg}(a) = \frac{1}{2a}\zeta_\omega(-1) = \frac{C_\zeta}{a}, \qquad C_\zeta = \frac{1}{2}\zeta_\omega(-1).$$

Since

$$V_{S^3}(a) = 2\pi^2 a^3,$$

the corresponding regularised density is

$$\rho_{\rm vac}^{\rm reg}(a) = \frac{C_\zeta}{2\pi^2 a^4}.$$

The a^{-4} law follows from scale covariance: energy scales as a^{-1} and volume as a^3.

B. Conformally shifted scalar reference spectrum

For a massless conformally shifted scalar on S^3, the relevant unit-sphere operator is $-\Delta_{S^3} + 1$, with

$$\nu_\ell = \ell + 1, \qquad d_\ell = (\ell+1)^2, \qquad \ell = 0, 1, 2, \ldots.$$

Thus

$$\zeta_\omega(s) = \sum_{\ell=0}^{\infty}(\ell+1)^2(\ell+1)^{-s} = \sum_{n=1}^{\infty} n^{2-s} = \zeta_{\rm R}(s-2).$$

Analytic continuation gives

$$\zeta_\omega(-1) = \zeta_{\rm R}(-3) = \frac{1}{120}, \qquad C_\zeta = \frac{1}{240},$$

and therefore

$$\rho_{\rm vac}^{\rm reg}(a) = \frac{1}{480\pi^2 a^4}.$$

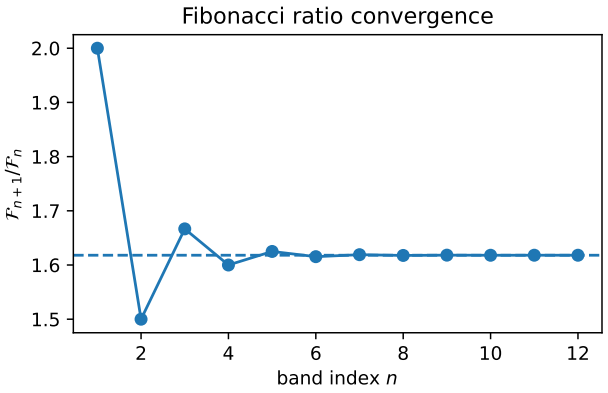


FIG. 1. The ratio $\mathcal{F}_{n+1}/\mathcal{F}_n$ converges to φ. The golden ratio is an asymptotic coding scale of the Fibonacci coarse-graining, not an independent physical coupling.

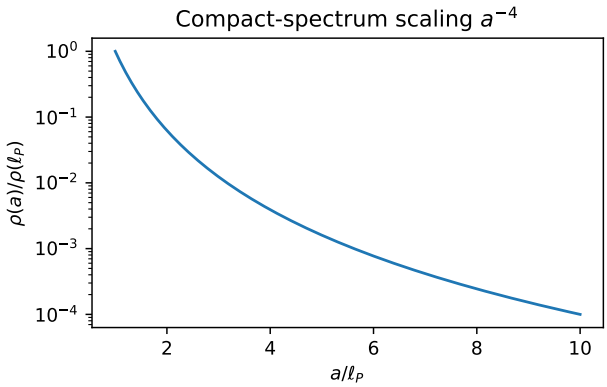


FIG. 2. Dimensionless compact-spectrum vacuum-density scale for the conformally shifted scalar reference spectrum. The exact scaling is a^{-4}.

Remark IV.1. The numerical coefficient and sign of a Casimir contribution depend on field content, spin structure, operator choice, subtraction prescription, and renormalisation convention. In semiclassical gravity the absolute vacuum energy also mixes with the renormalisation of the cosmological and gravitational couplings. The invariant statement used here is the compact spectral scaling and finiteness for $a \geq \ell_P$, not a direct prediction of the observed cosmological constant.

C. Fibonacci organization and the zeta sum

The Fibonacci bands B_n organize the same S^3 spectral tower. For a characteristic dimensionless frequency in a band one may use, for example,

$$\Omega_n = \sqrt{\nu_{\mathcal{F}_n}\nu_{\mathcal{F}_{n+1}-1}}, \qquad n \geq 1.$$

Then

$$\frac{\Omega_{n+1}}{\Omega_n} \longrightarrow \varphi \qquad (n \to \infty),$$

provided the underlying spectrum is asymptotically linear in ℓ, as it is for the conformal scalar frequencies.

A schematic band-counting zeta

$$\zeta_{\text{band}}(s) = \sum_{n\geq 1} |B_n| \Omega_n^{-s}$$

behaves asymptotically like a geometric series in φ^{1-s}, so it converges for $\operatorname{Re} s > 1$. If the full degeneracy of the three-dimensional spectrum is included, the large-band contribution scales as $\mathcal{F}_n^{3-s}$, and the usual convergence domain $\operatorname{Re} s > 3$ is recovered before analytic continuation.

Theorem IV.2 (Spectral-zeta vacuum-density scale)**.** *Let a compact S^3 spectral family satisfy $\omega_k(a) = \nu_k/a$, and let $\zeta_\omega(s) = \sum_k \nu_k^{-s}$ admit a finite analytic continuation at $s = -1$. Then*

$$E_{\text{vac}}^{\text{reg}}(a) = \frac{C_\zeta}{a},$$
$$\rho_{\text{vac}}^{\text{reg}}(a) = \frac{C_\zeta}{2\pi^2 a^4},$$
$$C_\zeta = \frac{1}{2}\zeta_\omega(-1).$$

In particular, $\rho_{\text{vac}}^{\text{reg}}$ is finite for every $a \geq \ell_P$. For the conformally shifted scalar reference spectrum,

$$\rho_{\text{vac}}^{\text{reg}}(a) = \frac{1}{480\pi^2 a^4}.$$

Proof. Scale covariance gives

$$E_{\text{vac}}^{\text{reg}}(a) = \frac{1}{2}\sum_k{}^{\text{reg}} \frac{\nu_k}{a} = \frac{1}{2a}\zeta_\omega(-1) = \frac{C_\zeta}{a}.$$

Dividing by $V_{S^3}(a) = 2\pi^2 a^3$ gives the stated density. Since $a \geq \ell_P > 0$ and $\zeta_\omega(-1)$ is finite by hypothesis, the density is finite at the Planck boundary. □

V. SELF-ADJOINT WHEELER–DEWITT BOUNDARY

The minisuperspace wave function is defined on the effective domain

$$a \in [\ell_P, \infty).$$

The sub-Planck interval $0 < a < \ell_P$ is not part of the Hilbert-space measure of this effective theory.

A schematic factor ordering for the Wheeler–DeWitt operator is

$$\hat{H} = -\frac{\hbar^2}{12\pi m_P^2}\frac{1}{a}\frac{\partial}{\partial a}\left(a\frac{\partial}{\partial a}\right) + \hat{H}_m + U_{\text{eff}}(a),$$

where $U_{\rm eff}$ contains curvature, matter, and regularised compact spectral contributions. The later argument requires not this precise potential but self-adjointness of the operator on the chosen domain.

The minisuperspace Hilbert space is

$$\mathcal{H}_{\rm mini} = L^2([\ell_P, \infty), a\, da),$$
$$\langle \Phi | \Psi \rangle = \int_{\ell_P}^{\infty} a\, \Phi^*(a) \Psi(a)\, da.$$

At the regular endpoint $a = \ell_P$, a local one-parameter self-adjoint boundary condition is

$$\cos\alpha\, \Psi(\ell_P) + \sin\alpha\, \ell_P \Psi'(\ell_P) = 0, \qquad \alpha \in [0, \pi).$$

At $a = \infty$ we assume either the usual limit-point case or, more generally, that the operator domain is chosen so that the boundary form vanishes there. Thus no probability flux is allowed through either endpoint of the effective half-line.

Theorem V.1 (Norm preservation in the effective minisuperspace domain)**.** *Assume that the Wheeler–DeWitt operator is given a self-adjoint extension on $L^2([\ell_P, \infty), a\, da)$, with vanishing boundary form at $a = \ell_P$ and at $a = \infty$. Then, for every relational or semiclassical clock parameter τ for which an effective Schrödinger form is valid,*

$$\frac{d}{d\tau} \|\Psi\|^2 = 0.$$

Equivalently, no probability is lost into the excluded sub-Planck region.

Proof. For the Sturm–Liouville differential expression

$$L\Psi = -\frac{1}{a}(a\Psi')',$$

the boundary form is proportional to

$$\left[a(\Phi^* \Psi' - \Phi'^* \Psi)\right]_{\ell_P}^{\infty}.$$

Self-adjointness is precisely the vanishing of this form on the operator domain. Taking $\Phi = \Psi$ gives zero probability flux at the endpoints. The usual Schrödinger conservation identity in any valid relational-time reconstruction therefore yields $d\|\Psi\|^2/d\tau = 0$. □

Corollary V.2 (Regular Planck boundary)**.** *The boundary condition permits $\Psi(\ell_P) \neq 0$ for generic Robin parameter α. The Planck boundary is therefore a regular endpoint of the effective quantum problem, not an absorbing singularity.*

A. WKB behaviour near the boundary

If a minisuperspace potential produces a classically forbidden interval adjacent to the Planck boundary, a WKB branch may have the schematic form

$$\Psi(a) \sim A \exp\left[-\frac{1}{\hbar}\int_{\ell_P}^{a} |p_a(a')|\, da'\right], \qquad a \geq \ell_P.$$

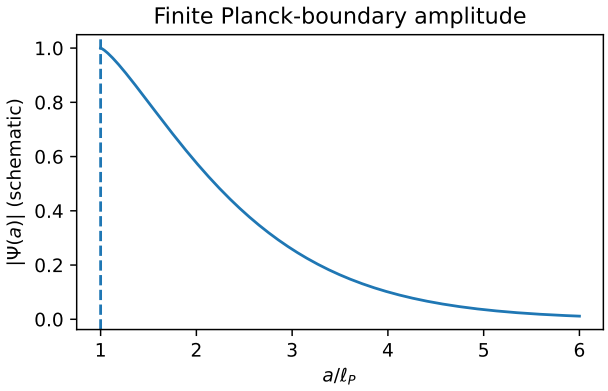


FIG. 3. Schematic finite-amplitude WKB profile on the effective domain $a \geq \ell_P$. No probability measure is assigned to $0 < a < \ell_P$.

This expression is used only on the physical domain. The value at $a = \ell_P$ is determined by the self-adjoint boundary condition and need not vanish.

Corollary V.3 (State-dependent Planck-scale correlations)**.** *In a compact S^3 universe, globally constrained quantum states need not factorise over arbitrary spatial partitions. Correlations between subregions may therefore persist to the Planck boundary. The corresponding entanglement entropy is state-dependent and is not fixed by topology alone.*

VI. NOETHER-II IDENTITIES ON CLOSED CAUCHY SLICES

Let the generally covariant action be written schematically as

$$S[g, \psi] = \int_{\mathcal{M}} d^4x\, \sqrt{-g}\, \mathcal{L}(g_{\mu\nu}, \psi, \nabla\psi).$$

Diffeomorphism invariance is a local gauge symmetry. By Noether's second theorem [14], it gives differential identities among the Euler–Lagrange expressions. It does not give, by itself, an independent global energy conservation law of the Noether-I type.

Theorem VI.1 (Noether-II identities and closed-slice boundary charges)**.** *Let a generally covariant theory be defined on a spacetime with closed Cauchy slice $\Sigma \simeq S^3$. Then:*

1. *Diffeomorphism invariance gives Noether-II differential identities among the Euler–Lagrange expressions. In Einstein gravity coupled to matter these include the contracted Bianchi identity and the corresponding covariant stress-energy conservation law on shell.*

2. *In Iyer–Wald covariant phase-space notation [16],*

$$J_\xi = \Theta(\phi; \mathcal{L}_\xi \phi) - \xi \cdot L, \qquad J_\xi \simeq dQ_\xi,$$

and the Hamiltonian variation associated with ξ is

$$\delta H_\xi = \int_{\partial\Sigma} (\delta Q_\xi - \xi \cdot \Theta)\,.$$

For the full closed slice, $\partial\Sigma = \varnothing$, so

$$\delta H_\xi = 0$$

for pure-gauge diffeomorphisms on the full slice.

3. *This full-slice statement does not exclude quasi-local charges on internal subdomains $D \subset \Sigma$ with $\partial D \neq \varnothing$.*

Proof. Part (1) is the standard Noether-II consequence of local diffeomorphism invariance. Part (2) is the Iyer–Wald variational identity evaluated on a Cauchy slice with empty boundary. Part (3) follows because an internal subdomain can have a boundary even when the whole Cauchy slice does not. □

Remark VI.2. For compact cosmology there is no ADM boundary at spatial infinity. Global "energy" is therefore not a boundary charge of the asymptotically flat type. The Hamiltonian content of general relativity in the closed case is carried by constraints and possible internal or quasi-local boundary terms.

A. Asymptotic Fibonacci scaling

The Fibonacci shift $n \mapsto n+1$ is discrete and only asymptotically represented by multiplication by φ. It is therefore not a continuous variational symmetry and does not fall under Noether's first or second theorem.

Proposition VI.3 (Asymptotic Fibonacci scaling law)**.** *For an exactly scale-covariant spectral family,*

$$\zeta_\omega(s; a) = a^s \zeta_\omega(s).$$

Consequently,

$$\zeta_\omega(s; \varphi a) = \varphi^s \zeta_\omega(s; a),$$

and at the regularised Casimir point,

$$E_{\rm vac}^{\rm reg}(\varphi a) = \varphi^{-1} E_{\rm vac}^{\rm reg}(a), \qquad \rho_{\rm vac}^{\rm reg}(\varphi a) = \varphi^{-4} \rho_{\rm vac}^{\rm reg}(a).$$

The Fibonacci interpretation of this rescaling is asymptotic in the band index.

Proof. The first identity is the definition of scale covariance. Substituting $a \mapsto \varphi a$ gives the second. The energy scaling follows at $s = -1$, and the density acquires the additional volume factor $(\varphi a)^{-3}$. □

VII. POSSIBLE OBSERVATIONAL SIGNATURES

The following items are qualitative targets, not predictions derived in this paper. A connection between Planck-boundary data and observations requires a separate perturbation-transfer model.

CMB power spectrum. The Fibonacci banding motivates searches for residual structure near multipoles

$$\ell = 2, 3, 5, 8, 13, 21, 34, 55, 89, \ldots.$$

The dipole $\ell = 1$ is not a clean primordial observable because it is dominated by the observer's motion. No universal amplitude is derived here.

Primordial power spectrum. A regular Planck boundary can modify admissible initial data for perturbation modes. The scale $1/\ell_P$ is ultraviolet, while low-ℓ CMB anomalies are infrared; relating them requires explicit inflationary or pre-inflationary transfer dynamics.

Dark-energy sector. The compact-spectrum term obtained above scales as a^{-4}. It therefore does not by itself mimic a constant late-time cosmological constant. DESI-class evolving-dark-energy fits are external phenomenological targets, not predictions of the present theorem [20].

Stochastic gravitational-wave background. A discrete compact S^3 tensor spectrum could motivate searches for resonance or modulation features in a stochastic background. The present-day frequencies would depend on the full expansion and transfer history and are not fixed here.

VIII. DISCUSSION

The model proves conditional statements inside a deliberately narrow domain. Its logical structure is

closed S^3 sector
$+\, a \geq \ell_P$
$+$ finite $H, \dot{H}$
$+$ self-adjoint WDW boundary
$+$ compact spectral zeta regularisation.

From these inputs one obtains finite FLRW curvature at the boundary, finite regularised compact-spectrum vacuum density, norm preservation in the minisuperspace Hilbert space, and the standard Noether-II identities of generally covariant dynamics.

A. Claims established here

1. Within the smooth closed connected orientable simply-connected spatial sector, the Poincaré–Perelman theorem together with Moise's theorem selects the standard S^3 sector.

TABLE I. Possible qualitative signatures. None is a definitive prediction of the present Planck-boundary theorem.

Observable	Possible qualitative signature	Required extra input
CMB low multipoles	Residual structure near Fibonacci multipoles $\ell \geq 2$	Perturbation transfer model
Primordial spectrum	Modified admissible initial data	Inflationary or pre-inflationary dynamics
Dark-energy sector	No direct prediction from the a^{-4} term	Additional late-time dynamics
Compact Casimir component	$\rho_{\rm vac}^{\rm reg} \propto a^{-4}$	Renormalisation and field content
SGWB	Possible discrete-spectrum modulation	Expansion and tensor-mode transfer history

2. Under finite Planck-boundary data, the closed-FLRW Kretschmann scalar obeys $K(t_P) \leq C/\ell_P^4$.

3. Zeta regularisation of a scale-covariant compact S^3 spectrum gives $\rho_{\rm vac}^{\rm reg} = C_\zeta/(2\pi^2 a^4)$, finite for $a \geq \ell_P$.

4. A self-adjoint Wheeler–DeWitt boundary condition prevents probability flux through $a = \ell_P$ in the effective minisuperspace description.

5. Diffeomorphism invariance gives Noether-II differential identities, and the full closed Cauchy slice has no Iyer–Wald boundary Hamiltonian variation.

B. Claims not established here

The following are not consequences of the present model:

1. a microscopic derivation of the Planck boundary;
2. a derivation of finite $H(t_P)$ and $\dot{H}(t_P)$;
3. an LQC-type bounce;
4. a prediction of the observed cosmological constant;
5. an exact Fibonacci spacetime symmetry;
6. a universal entanglement entropy fixed by topology;
7. a quantitative CMB or SGWB prediction.

C. Relation to loop quantum cosmology and CDT

The curvature result resembles loop quantum cosmology in its final boundedness statement, but the mechanism is different. LQC derives effective boundedness from polymer quantisation and a discrete geometric spectrum [17]. Here boundedness is conditional on a self-adjoint Planck boundary and finite semiclassical boundary data.

Causal dynamical triangulations supplies another independent example in which Planck-scale geometry may differ from smooth classical intuition [18]. The present Fibonacci coarse-graining is only an analogy at the level of effective spectral organization; it is not a derivation of CDT spectral dimensional flow.

IX. SUMMARY

The FB(S^3)R framework gives a rigorous conditional Planck-boundary construction for compact S^3 minisuperspace cosmology. Its main results are:

1. **Topological sector.** Assuming smooth closed connected orientable simple connectivity, the spatial slice is the standard S^3.

2. **Fibonacci coarse-graining.** The S^3 Laplacian spectrum may be organized into Fibonacci bands. The golden ratio enters only through the asymptotic limit $\mathcal{F}_{n+1}/\mathcal{F}_n \to \varphi$.

3. **Curvature bound.** For closed FLRW data with $a_{\rm eff}(t_P) = \alpha_P \ell_P$, $\alpha_P \geq 1$, and finite $H(t_P)$, $\dot{H}(t_P)$,
$$K(t_P) \leq \frac{C}{\ell_P^4} < \infty.$$

4. **Vacuum scale.** For a scale-covariant compact spectrum with finite $\zeta_\omega(-1)$,
$$\rho_{\rm vac}^{\rm reg}(a) = \frac{C_\zeta}{2\pi^2 a^4}.$$
For the conformally shifted scalar,
$$\rho_{\rm vac}^{\rm reg}(a) = \frac{1}{480\pi^2 a^4}.$$

5. **Self-adjoint boundary.** A self-adjoint Wheeler–DeWitt operator on $L^2([\ell_P, \infty), a\,da)$ gives zero probability flux through the Planck boundary.

6. **Noether-II structure.** Diffeomorphism invariance gives differential identities. On the full closed slice, Iyer–Wald boundary Hamiltonian variations vanish because $\partial\Sigma = \varnothing$, while internal quasi-local charges are not excluded.

Thus the model is mathematically coherent as a conditional effective Planck-boundary framework. Its strength is not that it solves quantum gravity, but that it states exactly which assumptions are sufficient for bounded curvature, compact-spectrum finiteness, and minisuperspace norm preservation.

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